

Sept 14th - finishing LM / Markov models
move tagging / HMM

Proof of MLE bigram example

training corpus $C = \{s_i\}_{i=1}^m$
 $s_i = x_{i,1}, \dots, x_{i,n_i}$ $n_i = |s_i|$

$$q_{ML} = \arg \max_{q \text{ p.d.}} \prod p(s_i)$$

prob of the entire corpus C

in def. of perplexity this
notion "prob of corpus" = $\prod$
of probs of sentences

$$= \arg \max_{q \text{ p.d.}} \log \prod p(s_i)$$

log is monotonic
if $a > b \geq 0$
 $\log(a) > \log(b)$

log preserves ordering

$$= \arg \max_{q \text{ p.d.}} \sum \log(p(s_i))$$

$$= \arg \max_{q \text{ p.d.}} \sum_{i=1}^m \sum_{j=1}^{n_i} \log q(x_{i,j} | x_{i,j-1})$$

use fact of bigram model

times that $q(v/w)$ shows up

$$= \arg \max_{q \text{ p.d.}} \sum_{\substack{w,v \\ \in V \cup \{\text{stop}\}}} c(v,w) \log q(w|v)$$

$$\arg \max_{q \text{ p.d.}} q(v/w) = \frac{c(w,v)}{c(w)}$$

some number $\neq 0$

$$\text{define } L = \sum_{i=1}^m \log(p(s_i)) + \sum_{v \in V} \lambda_v \left(\left[\sum_{\substack{w \\ \in V}} q(w|v) \right] - 1 \right)$$

$$q \text{ p.d.} \Leftrightarrow \forall v, w \ q(w/v) \geq 0 \quad \leftarrow \text{true bc } \log(q(w/v)) \text{ is in the equation}$$

$$\text{and } \forall v \sum_w q(w/v) = 1$$

$$\Leftrightarrow \forall v \left(\sum_w q(w/v) - 1 \right) = 0$$

For a p.d. q $h(q) = \sum \log(p(s_i))$

$$\arg \max_{q \text{ p.d.}} \sum \log(p(s_i)) = \arg \max_{q \text{ p.d.}} h(q)$$

but notice $\frac{\partial}{\partial \lambda_v} h(q, \lambda_v) \leftarrow L \text{ function of } q = \{q(w/v)\}_{w,v}, \lambda_v = \{\lambda_v\}_{v \in V}$

$$= \left[\sum_w q(w/v) \right] - 1$$

$$q \text{ is p.d.} \Leftrightarrow \forall v \in V, \frac{\partial}{\partial \lambda_v} h = 0$$

$$\therefore q \text{ s.t. for every } q(w/v) \text{ you have } \frac{\partial}{\partial q(w/v)} h = 0$$

$$\text{and every } \lambda_v, \frac{\partial}{\partial \lambda_v} h = 0$$

$$\Rightarrow q \text{ is a p.d. and maximizes } h \quad \leftarrow \text{(might be more complicated) SEE PIAZZA}$$

$$\Rightarrow q \text{ is p.d. that maximizes } \sum \log(p(s_i))$$

$$h = \sum_{w,v} c(v,w) \log q(w/v) + \sum_v \lambda_v \left(\sum_w q(w/v) - 1 \right)$$

$$\frac{\partial}{\partial q(w/v)} h = \frac{c(v,w)}{q(w/v)} + \lambda_v = 0$$

a single
variable

$$\Rightarrow q(w|v) = -\frac{c(v, w)}{\lambda_v}$$

since we set $\frac{\partial}{\partial q(w|v)} L = 0$, you get

$$q(w_1|v) = \frac{c(v, w_1)}{-\lambda_v} \quad q(w_2|v) = \frac{c(v, w_2)}{-\lambda_v}$$

$$\dots \quad q(w_{|v|}, v) = \frac{c(v, w_{|v|})}{-\lambda_v}$$

only way $q(w|v)$ is a p.d is if denominator

$$-\lambda_v = \sum_w c(v, w) = c(v)$$

$$\text{Indeed } \frac{\partial}{\partial \lambda_v} = \left[\sum_w q(w|v) \right] - 1 = 0$$

↓

$$\sum_w \frac{c(v, w)}{-\lambda_v} = 1$$

$$\text{optimal } \Rightarrow \lambda_v = -\sum_w c(v, w) = -c(v)$$

$$q(w|v) = \frac{c(v, w)}{c(v)} \quad \text{simultaneously for all } v, w$$

Lagrangian multiplier method

= dummy variables λ_v

changing gears: perplexity

reminder measure of how good a language model is

given test corpus T
perplexity $:= 2^{-l}$

$$Q = \frac{1}{M} \sum \log_2(p(s_i))$$

$$T = \{s_1, \dots, s_m\}$$

each sentence
 $|s_i| = n_i$

$$M = \sum n_i \quad \# \frac{\text{total words in}}{T}$$

$$2^{-L} = 2^{-\frac{1}{M} \sum \log_2 p(S_i)}$$

$$= \sqrt[M]{2^{-\sum \log_2 p(S_i)}}$$

$$= M \sqrt{2^{-\log_2(p(S_1))} \dots 2^{-\log_2(p(S_m))}}$$

$$= \sqrt[M]{\prod_{i=1}^m \left(\frac{1}{p(s_i)} \right)} = \frac{1}{\sqrt[M]{\prod_{i=1}^m p(s_i)}}$$

but for an n-gram model

$$= \sqrt[n]{\prod_{i=1}^m \prod_{j=1}^{n_i} q(x_{i,j} | x_{i,j-1}, x_{i,j-2}, \dots)}$$

↳ # terms in double product

$$\sum_{i=1}^m n_i = M$$

Def the geometric mean of M numbers
is $M\sqrt{x_1 \dots x_M}$

$$x_1, \dots, x_n \mapsto \sqrt[n]{x_1 \dots x_n}$$

log of this = arithmetic mean of the logs

Moral perplexity is the reciprocal of the geometric mean of the probs the model each word (each n-gram to be accurate in n-gram model case)

Cor if there is some n-gram $q(w/n, \dots) = 0$
 perplexity = ∞ (undefined)
 $\uparrow \lim_{q(w/\dots) \rightarrow 0} \text{perplexity} = \infty$

$$\underline{Q1} \quad p(x_1, \dots, x_n) = \gamma \times 0.5^n$$

if $x_n = \text{STOP}$

$$|V| = 3$$

valid ($p > 0$) sentences of length n

$$= \underbrace{3 \times \dots \times 3}_{n-1} = 3^{n-1}$$

p gives valid LM

$$\Leftrightarrow \sum_{s \in V^*} p(s) = 1$$

$$= \sum_{n=0}^{\infty} \sum p(s)$$

$$\begin{aligned}
& \sum_{n=1}^{\infty} \sum_{s \in |V|^{n-1} \times \{\text{STOP}\}} \gamma \left(\frac{1}{2}\right)^n \\
&= \sum_{n=1}^{\infty} 3^{n-1} \gamma \left(\frac{1}{2}\right)^n
\end{aligned}$$

γ can depend on n (γ_n) $\gamma_n = \frac{1}{3^{n-1}}$

$$= \sum_{n=1}^{\infty} \cancel{3^{n-1}} \frac{1}{\cancel{3^{n-1}}} \left(\frac{1}{2}\right)^n$$

$$= \sum \left(\frac{1}{2}\right)^n = 1$$

Perplexity problem (Q2)

max value of perplex $\rightarrow \infty$

$$n=1 \quad p(x) \rightarrow 0$$

$$\frac{1}{n} \log p(x) \rightarrow -\infty$$

$$2^{-\infty} \rightarrow \infty$$

min value of perplex = 1

$$n=1 \quad p(x) = 1$$

$$\frac{1}{n} \log_2 p(x) = 0$$

$$2^0 = 1$$

perp. $[1 - \frac{|V|}{\infty}]$

↑
uniform distribution
over all $|V|$ words
at every position